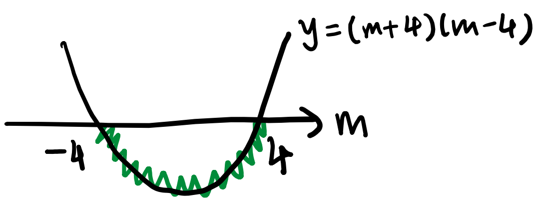
1 The equation of a curve is $xy = mx + 4$, where m is a constant.

(a) Find the set of values of m for which the curve and the line $x + y = 2m$ do not intersect. [4]

Solutions	**Comments**
$xy = mx + 4 - - - (1)$ $x + y = 2m$ $y = 2m - x - - - (2)$ Sub (2) into (1) : $x(2m - x) = mx + 4$ $-x^2 + 2mx = mx + 4$ $x^2 - mx + 4 = 0$ Discriminant < 0 $(-m)^2 - 4(1)(4) < 0$ $m^2 - 16 < 0$ $(m+4)(m-4) < 0$  Solution: $-4 < m < 4$	• Students should ensure clear presentation in their solutions. Instead of just writing $x^2 - mx + 4 = 0$ $(-m)^2 - 4(1)(4) < 0$, students should also write clearly discriminant < 0 before substitution of values • A common mistake is to take ±√ on both sides when solving quadratic inequality, which gives incorrect inequality solution: $m^2 - 16 < 0$ $m^2 < 16$ $m < \pm\sqrt{16}$ • Students should sketch the quadratic graph to identify the correct inequality solution easily

(b) Show that, for all values of m, the line $y = x + 1$ intersects the curve at two distinct points. [2]

Solutions	**Comments**
$xy = mx + 4 --- (1)$ $y = x + 1 --- (2)$ Sub (2) into (1): $x(x+1) = mx + 4$ $x^2 + x = mx + 4$ $x^2 + (1-m)x - 4 = 0$ Discriminant $= (1-m)^2 - 4(1)(-4)$ $= (1-m)^2 + 16$ For all values of m, $\underline{(1-m)^2 \geq 0}$ and hence $\underline{(1-m)^2 + 16 > 0}$. Since discriminant > 0, hence line and curve intersects at two distinct points.	• Students should recognise that the discriminant is already in completed square form. There is no need to expand the discriminant and complete the square again, which leads to redundant steps. • Two common mistakes in the explanation: 1. $(1 - m)^2 > 0$ (Not true because $(1 - m)^2$ can be 0 when $m = 1$) 2. Omitting the phrase "discriminant > 0" This phrase completes the explanation and provides the bridge between the expression $(1 - m)^2 + 16 > 0$ and why the line and curve intersects at two distinct points.

2 **(a)** Write down, and simplify, the first 3 terms in the expansion of $\left(1 - \frac{x}{2}\right)^8$ in ascending powers of x. [2]

Solutions	**Comments**
$\left(1-\frac{x}{2}\right)^8 = 1 + \binom{8}{1}\left(-\frac{x}{2}\right) + \binom{8}{2}\left(-\frac{x}{2}\right)^2 + \ldots$ $= 1 - 4x + 7x^2 + \ldots$	• Common mistakes: $1 + 4x + 7x^2 + \ldots$ Not including + … in the expansion

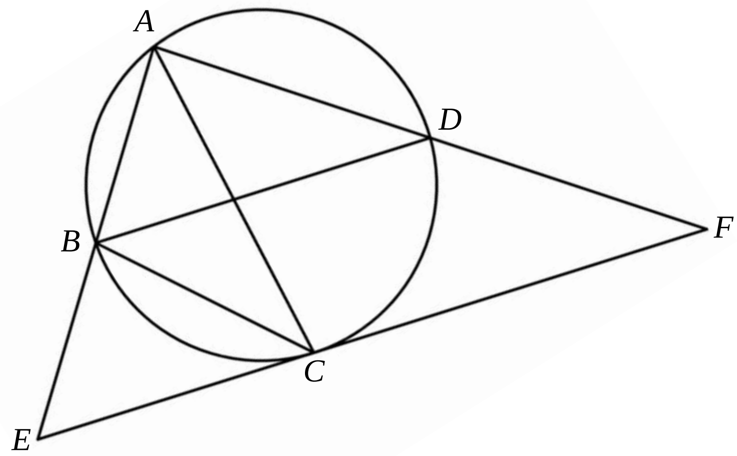
The expansion of $(1+x)^n\left(1-\frac{x}{2}\right)^8$, where n is a positive integer greater than 3, has no term in x^2.

(b) Find the coefficient of x in the expansion of $(1+x)^n\left(1-\frac{x}{2}\right)^8$. [5]

Solutions	Comments

$(1+x)^n\left(1-\frac{x}{2}\right)^8$
$=\left[1+\binom{n}{1}x+\binom{n}{2}x^2+\ldots\right](1-4x+7x^2+\ldots)$
$=\left(1+nx+\frac{n(n-1)}{2}x^2+\ldots\right)(1-4x+7x^2+\ldots)$
 	
Coefficient of x^2 $=-4n+\frac{n(n-1)}{2}+7$
 	
$-4n+\frac{n(n-1)}{2}+7=0$
$-8n+n^2-n+14=0$
$n^2-9n+14=0$
$(n-2)(n-7)=0$
 	
$n=2$ (reject) or $n=7$
 	
Coefficient of x
$=n-4$
$=3$
	• Students should recognise that this question is assessing the concept of selective expansion. Working should be shown clearly as to how the coefficient of x^2 in terms of n is obtained by writing out the expansion of $(1 + x)^n$ up to x^2 term clearly.
	
	• Common mistakes:
	  • Assuming $n = 4$ and expanding $(1 + x)^4$
	  • $\binom{n}{2}=\frac{n(n-1)(n-2)}{2}$
	  • Equating $\binom{n}{2}=0$ due to incorrect assumption that term in x^2 for $(1 + x)^n$ is zero.

3



The diagram shows a circle passing through the points A, B and D. The point E lies on AB produced such that $AB = BE$. The point C lies on the circle and EC is a tangent to the circle. The tangent EC meets AD produced at F and $AD = DF$. Prove that AC bisects angle EAF. [5]

Solutions	**Comments**
Since B is midpoint of AE and D is midpoint of AF, BD is parallel to EF by midpoint theorem. $\angle EAC = \angle ECB$ (angles in alt. segments) $= \angle CBD$ (alt. angles, $BD // EF$) $= \angle CAD$ (angles in same segment) Since $\angle EAC = \angle CAD$, AC bisects $\angle EAF$. **<u>OR</u>** Since B is midpoint of AE and D is midpoint of AF, BD is parallel to EF by midpoint theorem. Let X be intersection of AC and BD $\angle ACB = \angle ADX$ (angles in same segment) $\angle ABC = \angle ACF$ (angles in alt. segments) $= \angle AXD$ (corresp. angles, $BD // EF$) ΔBCA is similar to ΔXDA (AA Similarity Test) $\angle BAC = \angle XAD$ Therefore, AC bisects $\angle EAF$.	• Midpoint theorem consists of two conclusions (1) $EF = 2BD$ and (2) BD is parallel to EF. The 2nd conclusion was often omitted which was a critical step in the prove. • Common mistakes: • Assuming AC is a diameter of the circle and incorrect concluding that angle ACF is a right angle

4 **(a)** A viral outbreak occurs in a small town. The number of infected people, N, in the small town after t days can be modelled by the formula $N = ab^t$, where a and b are constants.
Values of N for different values of t have been collected.
Explain how a straight line graph can be drawn to represent the formula. State the axes of the graph and how the values of a and b can be obtained from the line. [4]

Solutions	**Comments**
$N = ab^t$ $\lg N = \lg a + (\lg b)t$ Using values of N and t, plot lg N against t and obtain straight line graph. lg N-intercept = lg a Gradient of line = lg b $a = 10^{\lg N\text{-intercept}}$, $b = 10^{\text{gradient}}$	• Common algebraic slips when taking logarithm on both sides include: • $\lg N = t \lg ab$ • $\lg N = \lg b + t \lg a$ • When linearising, use lg or ln only. Using $\log_b$ is not acceptable as b is unknown. Writing log without a base is also not acceptable. • Students need to understand the difference between unknown constants and variables. In this case, the formula $N = ab^t$ is a relationship between the **variables *N* and *t*** with a and b as constants to be found. In order to find this two constants, we linearise the formula into $\lg N = \lg a + (\lg b)t$ and draw a best fit line by plotting lg N against t (which are the new variables). A common mistake is to identify lg a or lg b as the axes to be drawn, which does not make sense since a and b are constants and so will lg a and lg b. • The method to find a and b needs to be stated clearly. A vague statement such as "a can be found from the Y-intercept" is insufficient explanation.

(b) In an experiment, a student recorded the time taken, t seconds, for an object to travel a distance of d metres. The table shows the recorded values of d and t.

d (metres)	2.00	4.00	6.00	8.00
t (seconds)	1.30	1.95	2.45	2.87

The student believed that d and t can be modelled by an equation of the form $d = pt + qt^2$, where p and q are constants.

(i) On the grid on page 7, draw a suitable straight line graph. [3]

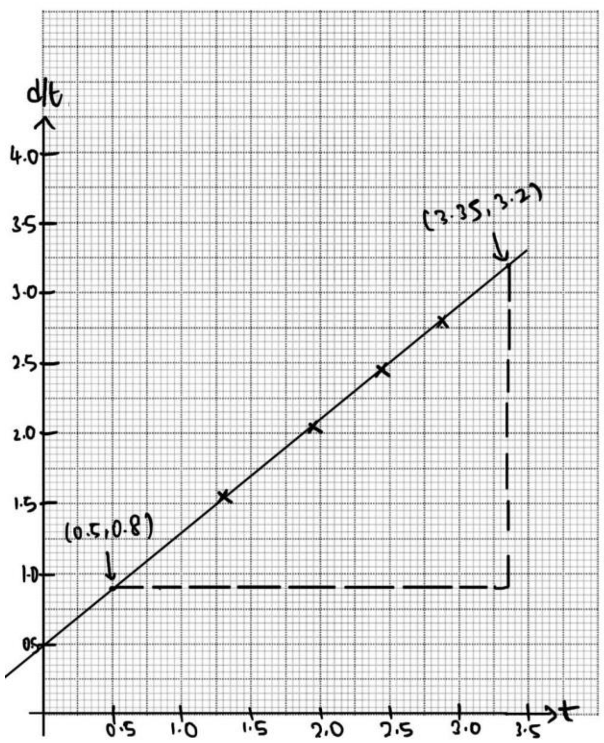
(ii) Use the graph to estimate the value of p and of q. [3]

Solutions

$d = pt + qt^2$

$\frac{d}{t} = p + qt$

t	1.30	1.95	2.45	2.87
d / t	1.54	2.05	2.45	2.79



$p = 0.5$

$q = \frac{3.2 - 0.9}{3.35 - 0.5}$

$= 0.807$ (3 s.f.)

Comments

- Students need to familarise themselves with linearising different types of non-linear equations. In (b)(i), the axes are not given and students are expected to recognise that $d = pt + qt^2$ is a non-linear equation in t that needs to be linearise by dividing by t.
- Table of values are to be tabulated in the space given or on the top right hand corner of the grid.
  The d / t values are to be corrected to 2 decimal places following the decimal places for d and t.
- (Issues related to choice of scale and plots) Looking at the values of t and d / t, choosing a scale such that each small square is 0.04 units will not be ideal (although is doable) as 3 out of the 4 t and d/t values cannot be plotted accurately and needs to be estimated.
- All readings to be read to half a small square if it is anywhere within the small square.
- Students should be reminded to avoid using broken axes for vertical axis as it may result in incorrect reading of the vertical axis-intercept.

5 The equation of a polynomial is given by $p(x) = 3x^3 + 2x^2 - 7x + 2$.

(a) Find the remainder when $p(x)$ is divided by $2x + 1$. [2]

Solutions	**Comments**
Remainder $= p(-0.5)$ $= 3(-0.5)^3 + 2(-0.5)^2 - 7(-0.5) + 2$ $= 5.625$	• Students who used long division to obtain the remainder should learn the remainder theorem method as it is more concise and efficient

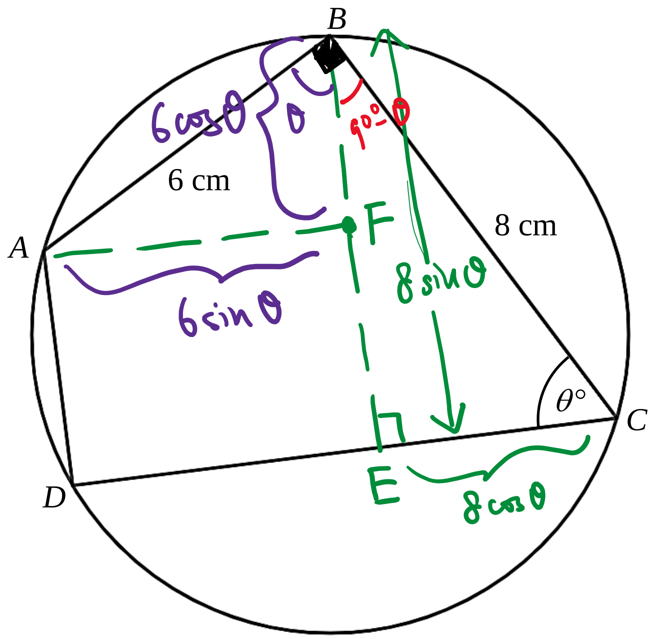
(b) Solve the equation $p(x) = 0$. [4]

Solutions	**Comments**
$p(1) = 3(1)^3 + 2(1)^2 - 7(1) + 2$ $= 0$ By Factor Theorem, $x - 1$ is a factor of $p(x)$. $3x^3 + 2x^2 - 7x + 2 = (x-1)(3x^2 + cx - 2)$	• Students should show clear substitution when using Factor Theorem in showing a linear expression is a factor of $p(x)$. (i.e. should not just write $p(1) = 0$.)
By comparing x^2, $-3 + c = 2$ $c = 5$	• Long division to obtain quadratic factor is accepted, although comparing coefficients will be a more efficient method.
$p(x) = 0$ $(x-1)(3x^2 + 5x - 2) = 0$ $(x-1)(3x-1)(x+2) = 0$ $x = 1$ or $x = \frac{1}{3}$ or $x = -2$	• Students are reminded to clearly write the equation to be solved instead of skipping steps from factorisation to the solution right away (i.e. following is not acceptable: $p(x) = (x-1)(3x^2 + 5x - 2)$ $= (x-1)(3x-1)(x+2)$ $x = 1$ or $x = \frac{1}{3}$ or $x = -2$ as $p(x) = 0$ is not seen.)

(c) Use your answer to **part (b)** to solve the equation $2(4^{3y})-7(4^{2y})+2(4^{y})+3=0$. [4]

Solutions	**Comments**
$2(64^{y})-7(4^{2y})+2(4^{y})+3=0$ $2(4^{3y})-7(4^{2y})+2(4^{y})+3=0$ $2-7(4^{-y})+2(4^{-2y})+3(4^{-3y})=0$ $3(4^{-y})^{3}+2(4^{-y})^{2}-7(4^{-y})+2=0$ $(4^{-y}-1)[3(4^{-y})-1](4^{-y}+2)=0$ $4^{-y}=1$ or $4^{-y}=\frac{1}{3}$ or $4^{-y}=-2$ (reject, $4^{-y}>0$) $y=0$ or $y=\dfrac{-\lg\frac{1}{3}}{\lg 4}$ $y=0$ or $y=0.792$ (3 s.f.)	• Students need use their **answer in part (b)**, which is the solutions of p(x) = 0 to solve the equation $2(4^{3y})-7(4^{2y})+2(4^{y})+3=0$. Using the method used in (b) again is NOT ACCEPTED. • Students are to show clearly how to compute the value $\log_4 3$ or $\log_4 \frac{1}{3}$ through change of base to lg or ln. Inputting it into the calculator but not showing working for it will result in loss of marks.

6



The diagram shows the design of a logo made up of a circle and a quadrilateral *ABCD*.
A, *B* and *C* are fixed points such that *AB* = 6 cm and *BC* = 8 cm.
AC is a diameter of the circle. *BC* makes an angle $\theta°$ with the line *CD*. The angle $\theta°$ can vary in such a way that the point *D* lies on the circle between points *A* and *C*.

(a) Show that the perimeter of *ABCD*, *P* cm, is given by $P = 14 + 14\sin\theta + 2\cos\theta$. [4]

Solutions	**Comments**
$\angle ABC = \angle ADC = 90°$ (right angle in semicircle) Let E be the foot of perpendicular from B to CD. Let F be the foot of perpendicular from A to BE. $BE = 8\sin\theta$ $EC = 8\cos\theta$ $\angle ABF = \theta$ and $\angle AFB = 90°$ $BF = 6\cos\theta$ $AF = 6\sin\theta$ $DC = AF + EC$ $= 6\sin\theta + 8\cos\theta$ $AD = BE - BF$ $= 8\sin\theta - 6\cos\theta$ $P = AB + BC + DC + AD$ $= 6 + 8 + 6\sin\theta + 8\cos\theta + 8\sin\theta - 6\cos\theta$ $= 14 + 14\sin\theta + 2\cos\theta$ (shown)	• Students are encouraged to use the diagram to indicate points and lengths. However, they still to show trigonometric working steps and show working for angles. • Students need to be familiar with breaking down any contextual *R*-Formula trigonometric problem with diagram given into right-angled triangles. A hint in the final form of *P* is the numbers 14 and 2, which can be derived from the addition and subtraction of trigonometric expressions involving 6 and 8. These trigonometric expressions will come from right-angled triangles with hypotenuse with 6 and 8.

(b) Express P in the form of $14 + R\sin(\theta + \alpha)$, where $R > 0$ and α is an acute angle. [4]

Solutions	**Comments**
$R = \sqrt{14^2 + 2^2}$ $= \sqrt{200}$ $\alpha = \tan^{-1}\left(\frac{2}{14}\right)$ $= 8.1301°$ $P = 14 + \sqrt{200}\sin\left(\theta + 8.1°\right)$ (1 d.p.)	• Students need to present the final form with the acute angle α to 1 decimal place as it is non-exact. The value for R in the final form can be presented exact (in surd form) or to 3 s.f.

(c) State the maximum value of P and find the corresponding value of θ. [3]

Solutions	**Comments**
Maximum $P = 14 + \sqrt{200}$ $= 28.1$ (3 s.f.) $\sin\left(\theta + 8.1301°\right) = 1$ $\theta + 8.1301° = 90°$ $\theta = 81.9°$ (1 d.p.)	• Common mistakes: • Max. value of $P = 1$ (confuse with max. value of $\sin\left(\theta + 8.1301°\right)$) • $\sin^{-1}(1) = 0°$ • Students should understand the question requirements which asks for **the** corresponding **value** (singular) of θ. Obtaining multiple solutions of θ definitely calls for rejecting of solutions and θ should not go above $180°$ (since D lies between B and C)

7 The equation of the tangent T to a circle at the point (6, –3) is $3y + 2x = 3$. The line with equation $4y + x + 20 = 0$ is a normal to the circle.

(a) Showing all working, find the equation of the circle. [7]

Solutions	**Comments**

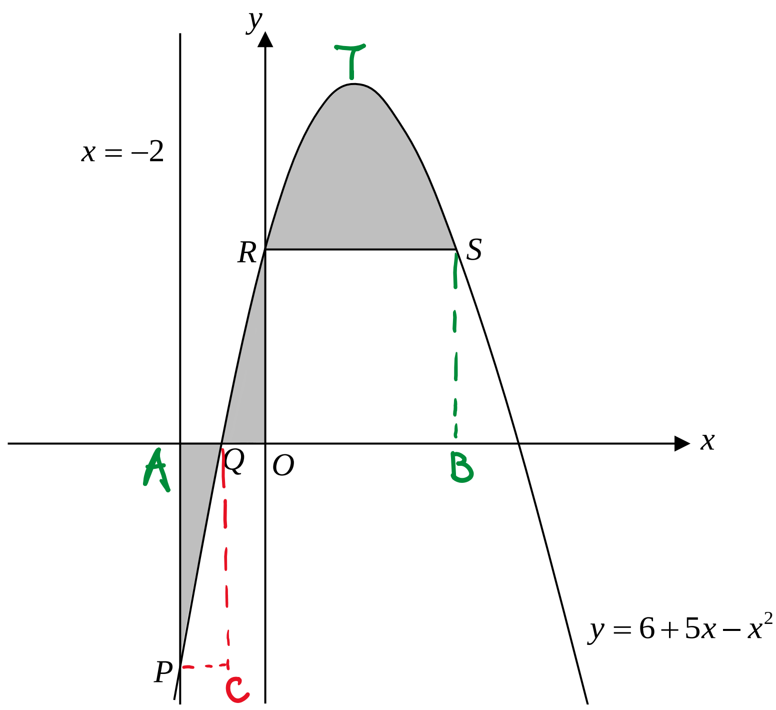
Let the centre of the circle be C and the point $(6,-3)$ be A
$3y+2x=3$
$y=-\frac{2}{3}x+1$
gradient of $T=-\frac{2}{3}$
gradient of $AC=-1\div-\frac{2}{3}$
$=1.5$
 	
equation of $AC:\frac{y+3}{x-6}=1.5$
$y=1.5x-12---(1)$
$4y+x+20=0---(2)$
Sub (1) into (2):
$4(1.5x-12)+x+20=0$
$6x-48+x+20=0$
$7x=28$
$x=4$
$y=-6$
$\therefore C(4,-6)$
 	
$\text{radius}^2=(6-4)^2+(-3+6)^2$
$=13$ units
 	
equation of circle :
$(x-4)^2+(y+6)^2=13$
	• Students are encouraged to sketch out a diagram to better visualise the question
	
	• For long questions that involves many steps and concepts, it is important to label your working clearly so that (1) you can follow your own working easily to check for errors (2) it allows the markers to identify potential error forward marks to give. In the case of unclear working due to lack of labelling, method marks may be deducted / not given.
	
	• Students need to be familiar with "normal to the circle". The normal to any curve is perpendicular to its tangent and for the case of a circle, the normal must be a diameter.
	
	• Common mistakes
	  • centre = intersection of the tangent and the normal
	  • $\text{radius}^2=(6-4)^2+(-3-6)^2$ [incorrect application of length formula]

P is the point on the circle that is nearest to the y-axis.

(b) Find the coordinates of the point at which the tangent at P and the tangent T intersects. [2]

Solutions	**Comments**
x-coordinate of $P = 4-\sqrt{13}$ Sub $x = 4-\sqrt{13}$ into equation of T : $y = -\frac{2}{3}(4-\sqrt{13})+1$ Coordinates of intersection $= (4-\sqrt{13}, -\frac{2}{3}(4-\sqrt{13})+1)$ $= (0.394, 0.737)$	• For presentation of answers, non-exact values (i.e. irrational values) should be left to 3 significant figures. This includes coordinates.

8



The diagram shows part of the curve of $y = 6 + 5x - x^2$. The line $x = -2$ intersects the curve at the point P. The curve meets the x-axis and y-axis at the points Q and R respectively.
The point S lies on the curve such that the line RS is parallel to the x-axis.

Find

(a) the coordinates of Q and S, [3]

Solutions	**Comments**
When $y = 0$, $0 = 6 + 5x - x^2$ $0 = (6 - x)(1 + x)$ $x = -1$ or $x = 6$ (reject) $\therefore Q(-1, 0)$ When $y = 6$, $6 + 5x - x^2 = 6$ $5x - x^2 = 0$ $x(5 - x) = 0$ $x = 0$ or $x = 5$ $\therefore S(5, 6)$	

(b) the total area of the shaded regions. [6]

Solutions	**Comments**
Area of left-most region $= -\int_{-2}^{-1} 6+5x-x^2 \text{ d}x$ $= -\left[6x+\frac{5x^2}{2}-\frac{x^3}{3}\right]_{-2}^{-1}$ $= -\left[-6+\frac{5(-1)^2}{2}-\frac{(-1)^3}{3}-\left(-12+\frac{5(-2)^2}{2}-\frac{(-2)^3}{3}\right)\right]$ $= 3\frac{5}{6}$ Area under curve from $x=-1$ to $x=5$ (*i.e.* middle shaded regions + area of *ORSB*) $= \int_{-1}^{5}(6+5x-x^2)\text{ d}x$ $= \left[6x+\frac{5x^2}{2}-\frac{x^3}{3}\right]_{-1}^{5}$ $= 6(5)+\frac{5(5)^2}{2}-\frac{(5)^3}{3}-\left(6(-1)+\frac{5(-1)^2}{2}-\frac{(-1)^3}{3}\right)$ $= 54$ Total area of shaded region $= 3\frac{5}{6}+54-(6)(5)$ $= 27\frac{5}{6}\text{ units}^2$	• Students need to understand that the integral with respect to x between $x = a$ and $x = b$ is used to find the area bounded by 4 boundaries: (1) the curve (2) the x-axis (3) the vertical line $x = a$ and (4) the vertical line $x = b$. It was a common mistake that students thought the integral $\int_{-2}^{-1} 6+5x-x^2 \text{ d}x$ gives the area *CPQ*, which is incorrect. • Students also need to identify when the integral will be negative. If students evaluate the area using the correct related integral and found the answer to be negative, they should simply add in the negative sign. Alternatively, just compute the integral and in the final step of adding or subtracting the areas, they must ensure they are just using the positive value of the integral. • An alternative method to compute the shaded area of the two middle regions is to calculate the integral Area of *QOR* + Area of *RTS* $= \int_{-1}^{0}(6+5x-x^2)\text{ d}x + \int_{0}^{5}(6+5x-x^2)-6\text{ d}x$

9 A particle, moving in a straight line, passes through a fixed point O. The velocity, v m/s, of the particle t seconds after passing through O, is given by $v = 20e^{-0.2t} - 2$.
The particle comes to instantaneous rest at the point P.
Find

(a) the value of t when the particle reaches P, [3]

Solutions	**Comments**
$20e^{-0.2t} - 2 = 0$ $e^{-0.2t} = \frac{1}{10}$ $-0.2t = \ln\left(\frac{1}{10}\right)$ $t = \frac{1}{-0.2}\left[-\ln 10\right]$ $= 5\ln 10$ $= 11.5129$ $= 11.5$ (3 s.f.)	

(b) the acceleration of the particle at P, [3]

Solutions	**Comments**
$a = (-0.2)(20e^{-0.2t})$ $= -4e^{-0.2t}$ When $t = 5 \ln 10$, $a = -4e^{-\ln 10}$ $= -0.4$ m/s^2	• Students who write substitute $t = 11.5129$ should actually substitute $t = 11.5129$ into the expression for acceleration a and obtain a non exact 3sf value of -0.400 m/s^2 . The only way to obtain exact -0.400 m/s^2 is to sub the exact value $t = 5 \ln 10$ • Students who use $t = 11.5$ will get the inaccurate value of -0.401 m/s^2 • Units for acceleration should be stated

(c) the distance travelled by the particle in the first 16 seconds. [6]

<table>
<tr><th>Solutions</th><th>Comments</th></tr>
<tr><td>

$s = \int (20e^{-0.2t} - 2)\, dt$

$= \dfrac{20}{-0.2} e^{-0.2t} - 2t + c$

$= -100e^{-0.2t} - 2t + c$

When $t = 0$, $s = 0$

$0 = -100e^{-0.2(0)} - 2(0) + c$

$c = 100$

$\therefore s = -100e^{-0.2t} - 2t + 100$

Distance $OP = -100e^{-\ln 10} - 10\ln 10 + 100$

$= (90 - 10\ln 10)$ m

Distance travelled in first 16 seconds

$= s\big|_{t=5ln10} + (s\big|_{t=5ln10} - s\big|_{t=16})$

$= 2 \times s\big|_{t=5ln10} - s\big|_{t=16}$

$= (180 - 20\ln 10) - (-100e^{-0.2(16)} - 2(16) + 100)$

$= 70.0$ m (3s.f.)

<u>OR</u>

Distance travelled in first 16 seconds

$= \left[\int_0^{5\ln 10} 20e^{-0.2t} - 2\, dt\right] - \left[\int_{5\ln 10}^{16} 20e^{-0.2t} - 2\, dt\right]$

(subtract as 2nd integral is < 0)

$= \left[-100e^{-0.2t} - 2t\right]_0^{5\ln 10} + \left[-100e^{-0.2t} - 2t\right]_{16}^{5\ln 10}$

$= 2[-100e^{-0.2\times 5ln10} - 2(5\ln 10)] - [-100e^{0} - 2(0)] - [-100e^{-0.2\times 16} - 2(16)]$

$= 70.0$ m (3 s.f.)

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- Common mistakes
  - $s = \int (20e^{-0.2t} - 2)\, dt$
$= \dfrac{20}{-0.2+1} e^{-0.2t} - 2t + c$
[confusing exponential integral with integral of power functions]
  - $c = 0$ or omitting c for indefinite integral
  - Distance travelled in first 16 seconds = Displacement at $t = 16$
  - Adding integral $\int_{5\ln 10}^{16} 20e^{-0.2t} - 2\, dt$ instead of subtracting [not recognising that integral is negative]

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</table>

10 **(a)** Differentiate $\ln(\sin x)$ with respect to x. [2]

Solutions	Comments
$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(\sin x)\right)=\frac{\cos x}{\sin x}$ $=\cot x$	

(b) Show that $\frac{\mathrm{d}}{\mathrm{d}x}(x\cot x)=\cot x-x\operatorname{cosec}^2 x$. [3]

Solutions	Comments
Method 1: Product rule on $x(\tan x)^{-1}$ $\frac{\mathrm{d}}{\mathrm{d}x}(x\cot x)=\frac{\mathrm{d}}{\mathrm{d}x}[x(\tan x)^{-1}]$ $=(\tan x)^{-1}+x(-1)(\tan x)^{-2}(\sec^2 x)$ $=\cot x-x\left(\frac{\sec^2 x}{\tan^2 x}\right)$ $=\cot x-x\left[\frac{1}{\cos^2 x}\times\frac{\cos^2 x}{\sin^2 x}\right]$ $=\cot x-\frac{x}{\sin^2 x}$ $=\cot x-x\operatorname{cosec}^2 x$ Method 2: Product & Quotient rule on $x\left(\frac{\cos x}{\sin x}\right)$ $\frac{\mathrm{d}}{\mathrm{d}x}(x\cot x)=\frac{\mathrm{d}}{\mathrm{d}x}[x\left(\frac{\cos x}{\sin x}\right)]$ $=\left(\frac{\cos x}{\sin x}\right)+x\left(\frac{\sin x(-\sin x)-\cos x(\cos x)}{\sin^2 x}\right)$ $=\cot x+x\left(\frac{-\sin^2 x-\cos^2 x}{\sin^2 x}\right)$ $=\cot x+x\left(\frac{-1}{\sin^2 x}\right)$ $=\cot x-\frac{x}{\sin^2 x}$ $=\cot x-x\operatorname{cosec}^2 x$	• Students are to show clear application of trigonometric identities without omitting any essential steps. After writing out the proof, it will be good to briefly look through to ensure no essential steps were skipped.

(c) Using the results from **parts (a)** and **(b)**, show that

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} x \operatorname{cosec}^2 x \, dx = \frac{(2\sqrt{3}-3)\pi}{12} + \frac{1}{2}\ln 2 .$$ [6]

Solutions	**Comments**
$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} x \operatorname{cosec}^2 x \, dx$ $= -\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (-x \operatorname{cosec}^2 x)\, dx$ $= -\left[\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\cot x - x \operatorname{cosec}^2 x)\, dx - \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot x \, dx\right]$ $= \left[-x\cot x\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot x \, dx$ $= \left[-x\cot x + \ln(\sin x)\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$ $= \left(-\frac{\pi}{4}\right)\cot\left(\frac{\pi}{4}\right) + \ln\left(\sin\frac{\pi}{4}\right) + \left(\frac{\pi}{6}\right)\cot\left(\frac{\pi}{6}\right) - \ln\left(\sin\frac{\pi}{6}\right)$ $= -\frac{\pi}{4} + \ln\left(\frac{1}{\sqrt{2}}\right) + \frac{\pi}{6}\left(\sqrt{3}\right) - \ln\left(\frac{1}{2}\right)$ $= \frac{2\sqrt{3}\pi}{12} - \frac{3\pi}{12} - \ln\left(\sqrt{2}\right) + \ln 2$ $= \frac{(2\sqrt{3}-3)\pi}{12} + \frac{1}{2}\ln 2$ **<u>OR</u>** From (b), $\int (\cot x - x \operatorname{cosec}^2 x)\, dx = x\cot x + c$ $\int \cot x \, dx - \int x \operatorname{cosec}^2 x \, dx = x\cot x + c$ $\int x \operatorname{cosec}^2 x \, dx = \int \cot x \, dx - x\cot x - c$ $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} x \operatorname{cosec}^2 x \, dx$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot x \, dx - \left[x\cot x\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$ $= \left[\ln(\sin x) - x\cot x\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$ $= \left(-\frac{\pi}{4}\right)\cot\left(\frac{\pi}{4}\right) + \ln\left(\sin\frac{\pi}{4}\right) + \left(\frac{\pi}{6}\right)\cot\left(\frac{\pi}{6}\right) - \ln\left(\sin\frac{\pi}{6}\right)$ $= -\frac{\pi}{4} + \ln\left(\frac{1}{\sqrt{2}}\right) + \frac{\pi}{6}\left(\sqrt{3}\right) - \ln\left(\frac{1}{2}\right)$ $= \frac{2\sqrt{3}\pi}{12} - \frac{3\pi}{12} - \ln\left(\sqrt{2}\right) + \ln 2$ $= \frac{(2\sqrt{3}-3)\pi}{12} + \frac{1}{2}\ln 2$	• Students need to recognise that the required integral comes from reverse the differentiation process from (b) [look at method **<u>OR</u>**] • Students need to write out integral notations clearly. The dx should be present and any definite integrals should one upper and one lower limit. If indefinite integrals are computed, the integration constant c is expected to be written down. • When integrating two or more algebraic terms, it is a good habit to put a bracket around the entire expression for clearer presentation: $\int (\cot x - x \operatorname{cosec}^2 x)\, dx$ is preferred to $\int \cot x - x \operatorname{cosec}^2 x \, dx$ • Clear presentation to be shown how $\ln\left(\frac{1}{\sqrt{2}}\right) - \ln\left(\frac{1}{2}\right)$ can be simplified to $\frac{1}{2}\ln 2$ using logarithm laws